SQL MINI PROJECT REPORT

AMAZON FRESH ANALYTICS DATABASE

Project Details	**Information**
Project Title	Amazon Fresh Analytics Database
Database	Amazon
Tools used	MySQL Workbench
Project Type	SQL Mini Project
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AMAZON FRESH ANALYTICS – SQL MINI PROJECT REPORT

1. INTRODUCTION

1.1 Project Overview

The Amazon Fresh Analytics SQL Project is a relational database project developed using MySQL. The project is designed around an online grocery and retail business environment where large amounts of customer, product, supplier, order and review information need to be stored and managed efficiently.

In an e-commerce business, data is generated continuously through customer registrations, product additions, orders, payments, inventory updates and customer reviews.

In this project, an Amazon database was created using MySQL Workbench. Different tables were used to represent important business entities such as Customers, Products, Suppliers, Orders, Order Details and Reviews. Relationships between these tables were established using primary and foreign keys.

1.2 Purpose of the Project

The main purpose of this project is to gain practical knowledge of SQL and relational database management by working with an e-commerce dataset.

The project focuses on:

Creating and managing a relational database.

Understanding table structures and relationships.

Maintaining data integrity.

Performing data manipulation.

Filtering and analyzing records.

Combining information from multiple tables.

Performing customer and product analysis.

Applying normalization concepts.

Extracting useful business insights.

1.3 Tools Used

The project was developed using MySQL and MySQL Workbench. MySQL was used for creating the database, managing tables and executing SQL operations. MySQL Workbench was also used to create and visualize the Entity Relationship Diagram

2. PROJECT OBJECTIVES AND SCOPE

The main objective of the project is to develop practical knowledge of SQL by implementing a relational database based on an Amazon Fresh business scenario.

The project objectives include:

To create an Amazon database using MySQL.

To understand the structure of relational databases.

To identify different entities and their attributes.

To create relationships between different tables.

To implement Primary Keys and Foreign Keys.

To retrieve specific information using SQL filtering.

To perform INSERT, UPDATE and DELETE operations.

To apply different SQL constraints.

To analyze customer and product information.

To perform aggregation and grouping.

To combine multiple tables using joins.

To perform sales and revenue analysis.

To identify high-value customers.

To analyze supplier stock information.

To identify Prime Member concentration.

To apply database normalization.

To extract useful business insights.

3. DATABASE TABLES

The project contains the following major tables:

Customers

Stores customer details such as Customer ID, Name, Age, Gender, City, State, Country, Signup Date and Prime Member status.

Products

Stores Product ID, Product Name, Category, Subcategory, Price, Stock Quantity and Supplier information.

Suppliers

Contains information about suppliers who provide products.

Orders

Stores order-related information such as Order ID, Customer ID, Order Date, Order Amount, Delivery Fee and Discount.

Order Details

Contains product-level information for each order, including Quantity, Unit Price and Discount.

Reviews

Stores customer reviews and product ratings.

4. DATABASE DESIGN AND ER DIAGRAM

The database was designed using a relational structure where information is divided into different tables.

Primary Keys and Foreign Keys were used to connect the tables and maintain data relationships.

The major relationships include:

Customers and Orders

Orders and Order Details

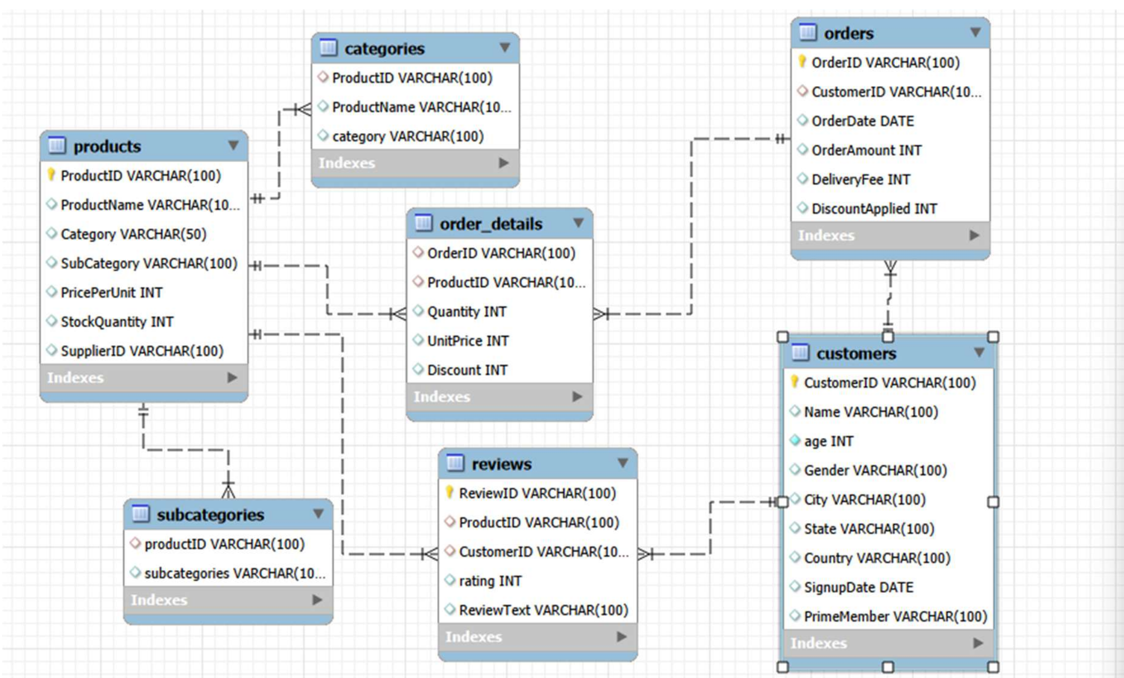
Products and Order Details

Products and Reviews

Customers and Reviews

Suppliers and Products

An ER Diagram was created using MySQL Workbench to represent these relationships visually.



5. SQL OPERATIONS AND CONSTRAINTS

5.1 Data Retrieval

The project includes different data retrieval tasks using filtering conditions.

Customer records were retrieved based on a specific city, while products were filtered based on their category.

These operations demonstrate how SQL can be used to retrieve only the required information from a larger dataset.

5.2 INSERT Operation

New product records were added to the Products table.

This demonstrates how new products can be introduced into an existing database.

The inserted product information included product identification, product name, category, subcategory, price, stock quantity and supplier information.

5.3 UPDATE Operation

The stock quantity of a particular product was updated.

This represents a real-world inventory management situation where the available stock of a product changes over time.

5.4 DELETE Operation

A supplier record was identified and deleted as part of the assigned task.

This demonstrates how unwanted or outdated records can be removed from a database.

5.5 Constraints

Several SQL constraints were implemented to maintain data accuracy.

Primary Key: Used to uniquely identify records.

Foreign Key: Used to establish relationships between tables.

NOT NULL: Used to ensure required fields are not left empty.

UNIQUE: Used to prevent duplicate values.

CHECK: Used to ensure that review ratings remain within the valid range of 1 to 5.

DEFAULT: Used to provide a default value for Prime Member status.

6. DATA ANALYSIS USING SQL

6.1 Filtering and Grouping

The project uses filtering and grouping techniques to analyze the available data.

Orders were filtered based on dates to identify transactions within a particular period.

Products were grouped based on Product ID to calculate sales-related values.

Customers were grouped to calculate individual customer spending.

6.2 Aggregate Functions

Aggregate functions were used to summarize data.

SUM was used to calculate total spending, revenue and stock quantities.

COUNT was used to count customers, orders and product occurrences.

AVG was used to calculate average product ratings.

These functions convert individual records into useful business summaries.

6.3 GROUP BY

GROUP BY was used to organize records based on common attributes.

For example, customers can be grouped based on Customer ID and products can be grouped based on Product ID.

Grouping is particularly useful when combined with aggregate functions.

6.4 HAVING

The HAVING condition was used to filter grouped results.

For example, products with an average rating above a specified value were identified.

This helps focus the analysis only on records that meet the required business condition.

6.5 ORDER BY

ORDER BY was used to arrange results based on specific values.

Products can be arranged from highest to lowest sales, customers can be arranged based on spending and suppliers can be arranged based on stock quantities.

6.6 Business Importance

These analytical techniques allow the database to answer practical business questions.

Instead of simply displaying raw records, the data can be summarized and compared to identify trends and important business segments.

7. CUSTOMER, SALES AND SUPPLIER ANALYSIS

7.1 Customer Spending Analysis

Customer spending was analyzed using order information.

The total amount spent by each customer was calculated and customers were ranked according to their spending.

This helps identify customers who contribute significantly to the business revenue.

7.2 High-Value Customers

Customers exceeding the specified spending level were identified as high-value customers.

These customers can be important for customer retention because they contribute a larger amount of revenue.

Businesses can provide loyalty benefits, personalized offers and special promotions to such customers.

7.3 Order Frequency

Customer order frequency was analyzed by counting the number of orders placed by customers.

Customers who place orders frequently can be considered more engaged with the platform.

This information can support customer loyalty and retention strategies.

7.4 Revenue Analysis

Order and Order Details information was combined to calculate revenue.

The quantity and unit price of products were considered when calculating product-level revenue.

This provides a better understanding of how individual products and orders contribute to overall sales.

7.5 Supplier Stock Analysis

Supplier and product information were combined to analyze stock quantities.

The total stock associated with each supplier was calculated and suppliers could be compared based on their inventory contribution.

This analysis can help the business understand supplier dependency and support inventory planning.

8. PRODUCT, PRIME MEMBER AND ORDER ANALYSIS

8.1 Product Sales Analysis

Products were analyzed based on their sales revenue.

The quantity sold and unit price were considered to determine the revenue generated by different products.

The highest-performing products can be identified from this analysis.

8.2 Product Rating Analysis

Customer reviews were analyzed to calculate average product ratings.

Products with higher average ratings can be identified and compared.

This can help understand customer satisfaction and product performance.

8.3 Customers Without Orders

The database was also analyzed to identify customers who have not placed any orders.

This information can be useful for identifying potential customers who have registered but have not yet purchased products.

Such customers can be targeted through promotional campaigns and first-order offers.

8.4 Prime Member Analysis

Prime Members were analyzed based on their city.

This helps identify cities where the number of Prime Members is higher.

Geographical membership information can be useful for regional marketing and customer retention strategies.

8.5 Frequently Ordered Categories

Product and order information was combined to understand which categories occur frequently in customer orders.

This can help the business understand customer preferences and demand patterns.

It can also support inventory planning and product assortment decisions.

9. NORMALIZATION AND ADVANCED SQL ANALYSIS

9.1 Normalization

Normalization was performed as part of the project to improve database organization.

The category and subcategory information associated with products was separated into dedicated tables.

The additional tables created were Categories and Subcategories.

This reduces unnecessary repetition and makes category-related information easier to maintain.

9.2 Benefits of Normalization

Normalization provides several advantages:

Reduces data redundancy.

Improves data consistency.

Makes database maintenance easier.

Provides better organization.

Reduces repeated information.

Makes relationships clearer.

9.3 Subquery / Advanced Analysis

The project also includes advanced SQL analysis for identifying top-performing products and analyzing customer-related information.

The analysis was used to identify products based on their sales revenue and customers based on their order activity.

9.4 Ranking

Ranking was applied to customer spending to identify high-value customers.

Ranking provides a simple way to compare customers based on their contribution to overall business revenue.

9.5 Business Importance

These advanced analytical techniques demonstrate how SQL can be used for decision-making rather than only basic data retrieval.

The results can support customer management, product planning, inventory decisions and marketing activities.

10.Amazon Sales & Customer Insights Dashboard

11. KEY FINDINGS AND CONCLUSION

KEY FINDINGS

Customer Insights

The database can be used to identify customers based on their spending, order frequency, location and Prime Member status.

Product Insights

Products can be analyzed based on their category, sales revenue, ratings and stock availability.

Supplier Insights

Supplier information can be connected with product inventory to understand supplier-wise stock contribution.

Sales Insights

Order and order-detail information can be combined to calculate revenue and identify high-performing products.

Review Insights

Customer reviews can be analyzed to identify products with higher average ratings.

Geographic Insights

Customer location and Prime membership information can be used to understand regional customer concentration.

CONCLUSION

The Amazon Fresh Analytics SQL Project provided practical experience in relational database design and SQL-based data analysis.

The project covered database creation, table relationships, Primary Keys, Foreign Keys, constraints, data manipulation, filtering, aggregation, joins, ranking, normalization and business analysis.

The database was used to analyze customers, products, suppliers, orders and reviews. These analyses demonstrate how SQL can convert raw business data into meaningful information.

Overall, the project helped develop practical knowledge of MySQL, relational database management, SQL analysis and business-oriented data interpretation.